

max Total Reward
min Ave Regret

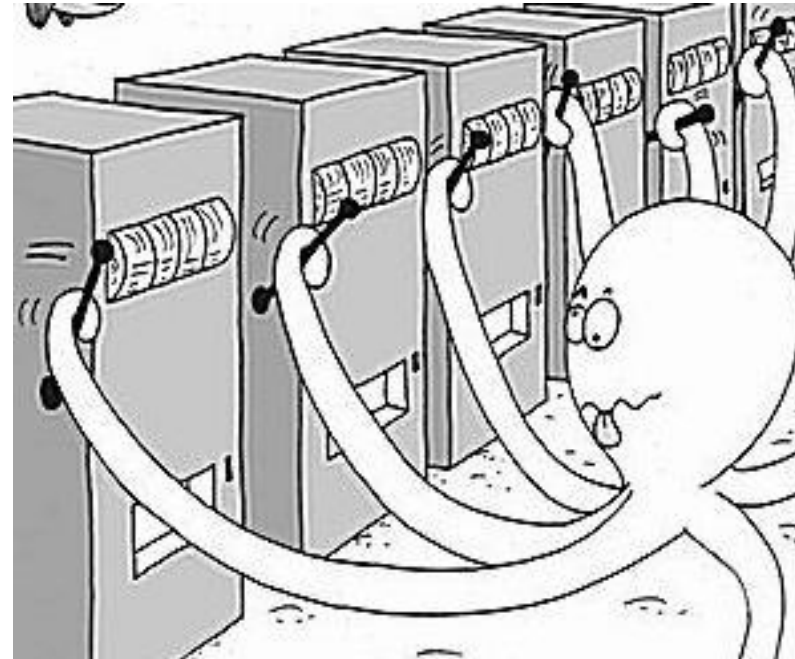
$$A = \{1, \dots, n\}$$

$v_a^* \sim \text{Distribution}$

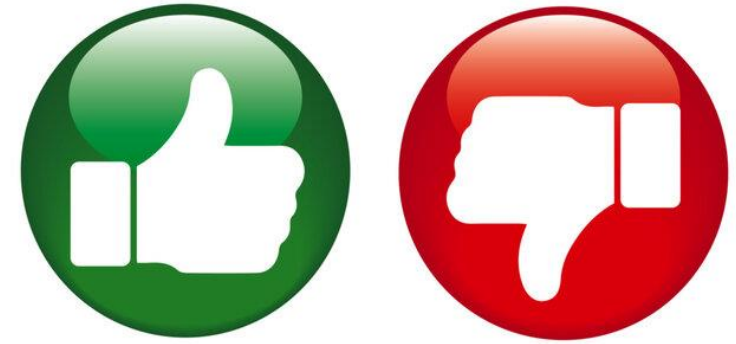
Multi-Armed Bandits

~~$D = \{(s, a) \dots (s', a)\}$~~

Daniel Brown



Evaluative feedback



REPORT CARD	
Reading	B
Writing	C-
Mathematics	D
Science	C-
History	B+
Art	B-
P.E.	B

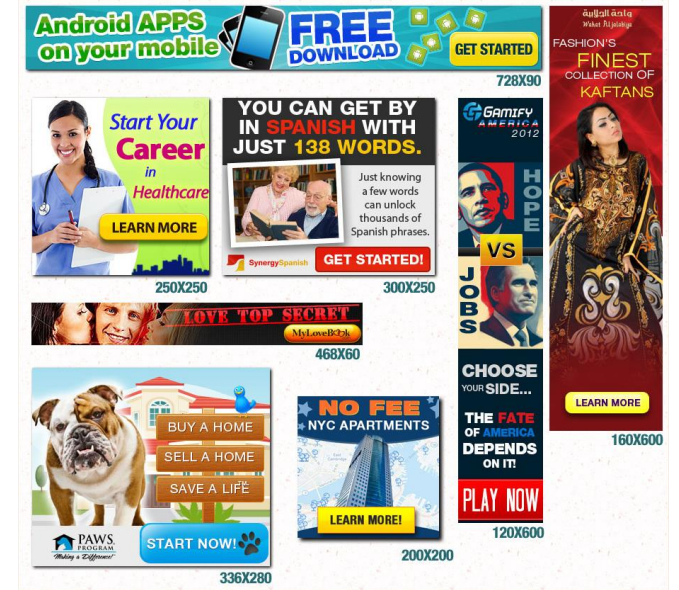


Applications

$A^?$ $R^?$
 $\pi \rightarrow A$

- Online Advertising and Recommendation
- Clinical Trials
- Robotics
- Dynamic Pricing
- Search Engine Optimization
- Education and Learning Platforms

$MAB \rightarrow$ Contextual Bandit

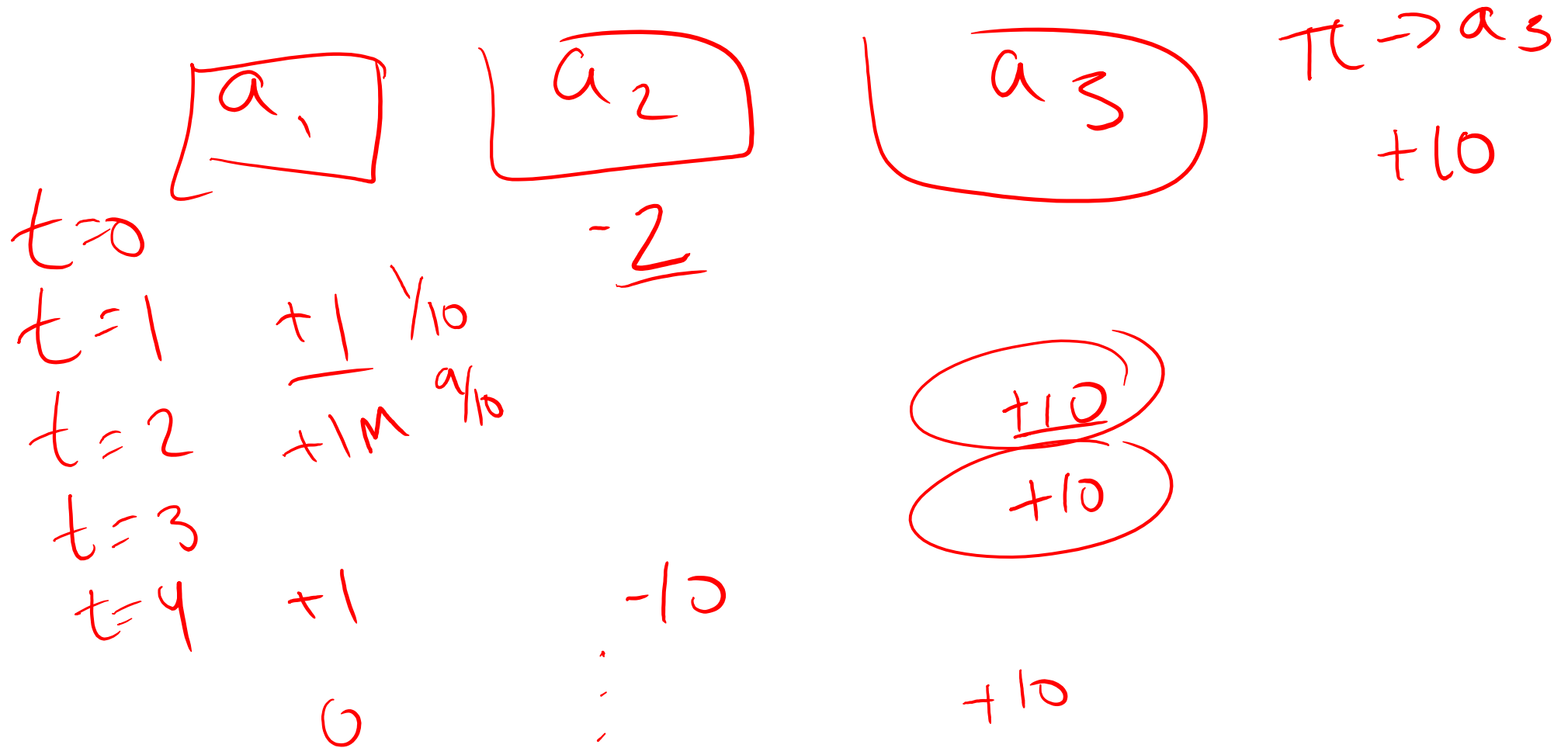


Problem formalism

- ^{Actions} Arms $\mathcal{A} = \{a_1, \dots, a_k\}$
 - Each arm is associated with an unknown reward distribution
- Rewards $r_t(a_i)$ - $\mathbb{E}[r(a_i)] = \mu_{a_i}$
- Possible Goals
 - Maximize cumulative reward (Minimize regret)
 - Best arm identification
- Standard Assumptions
 - Independence: Rewards from each arm are independent
 - Stationarity: Reward distributions don't change over time

$\hookrightarrow$ if visited $\epsilon \triangleq 0.1$

How should we solve this problem?

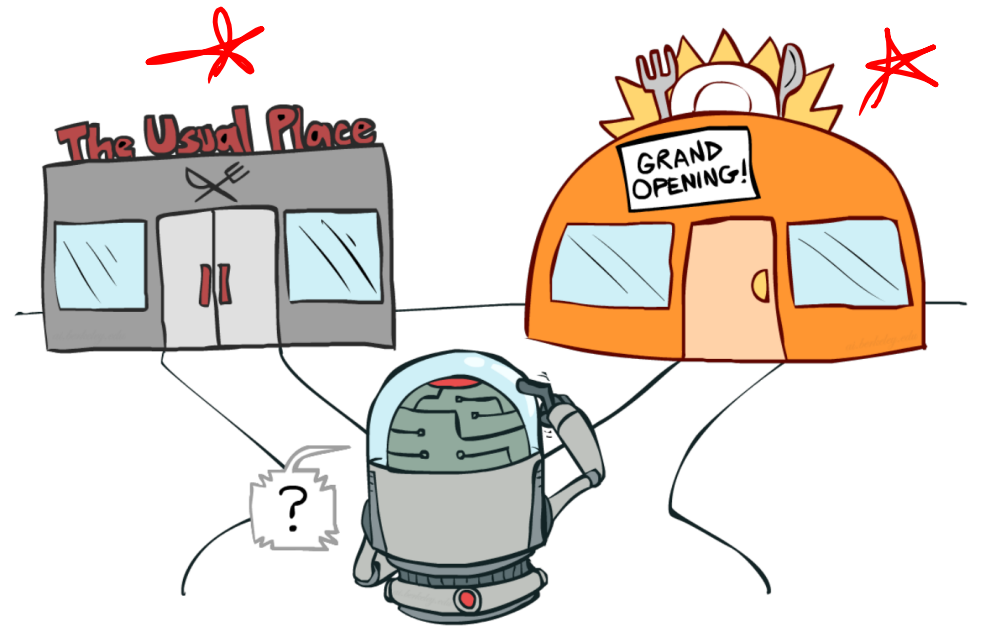


Random

Big Regret

Greedy

Exploration



ϵ -Greedy

$\epsilon \in (0, 1)$

e.g. 0.1, 0.05, 0.01
* anneal/reduce/decay
 ϵ over time

loop: rand $\in [0, 1]$

random (if rand $< \epsilon$:
explore action at random

greedy (else:
act greedy

2.3 The 10-armed Testbed

To roughly assess the relative effectiveness of the greedy and ε -greedy action-value methods, we compared them numerically on a suite of test problems. This was a set of 2000 randomly generated k -armed bandit problems with $k = 10$. For each bandit problem, such as the one shown in Figure 2.1, the action values, $q_*(a)$, $a = 1, \dots, 10$,

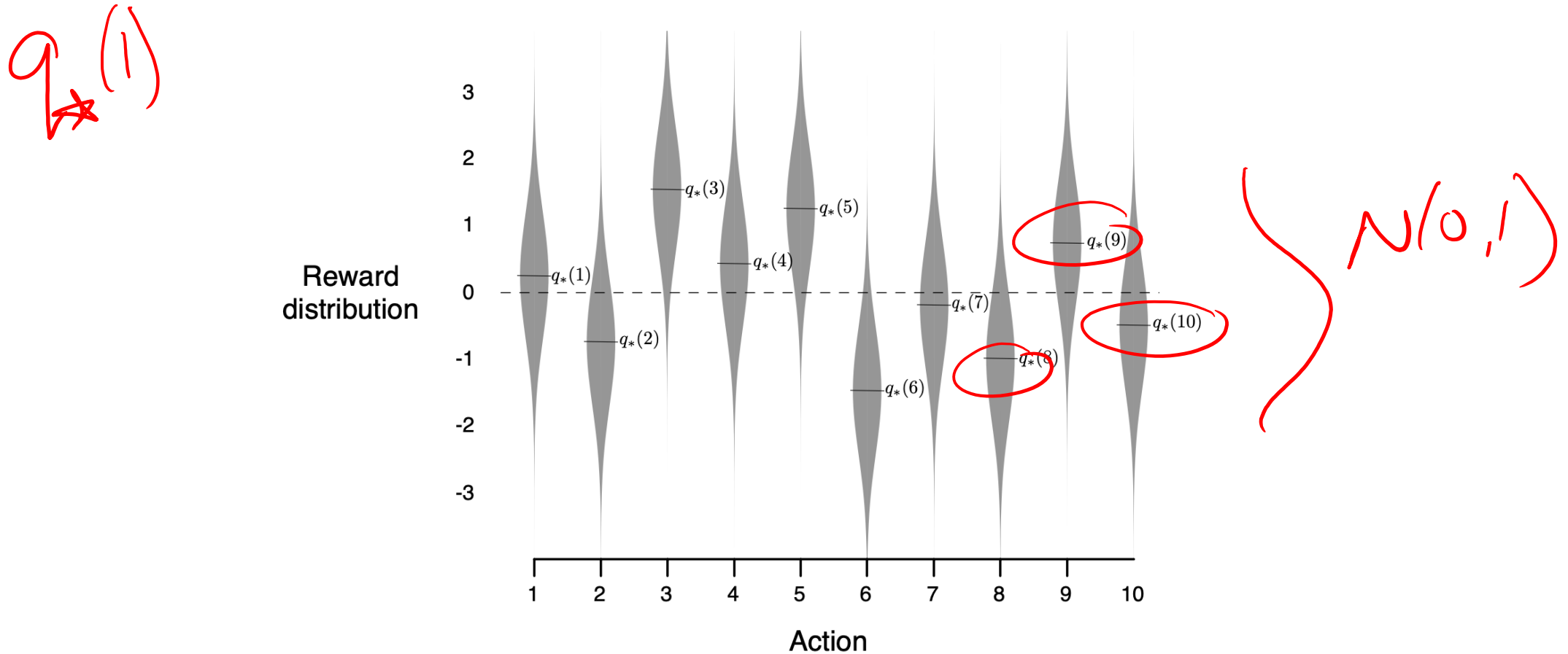


Figure 2.1: An example bandit problem from the 10-armed testbed. The true value $q_*(a)$ of each of the ten actions was selected according to a normal distribution with mean zero and unit variance, and then the actual rewards were selected according to a mean $q_*(a)$ unit variance normal distribution, as suggested by these gray distributions.

Sutton/Barto figure

- 10 arms
- Each arm has stochastic reward
 $r \sim N(Q^*(a), 1)$
- Averaged over 2000 bandit problems where each problem starts with $Q^*(a) \sim N(0, 1)$ for all a

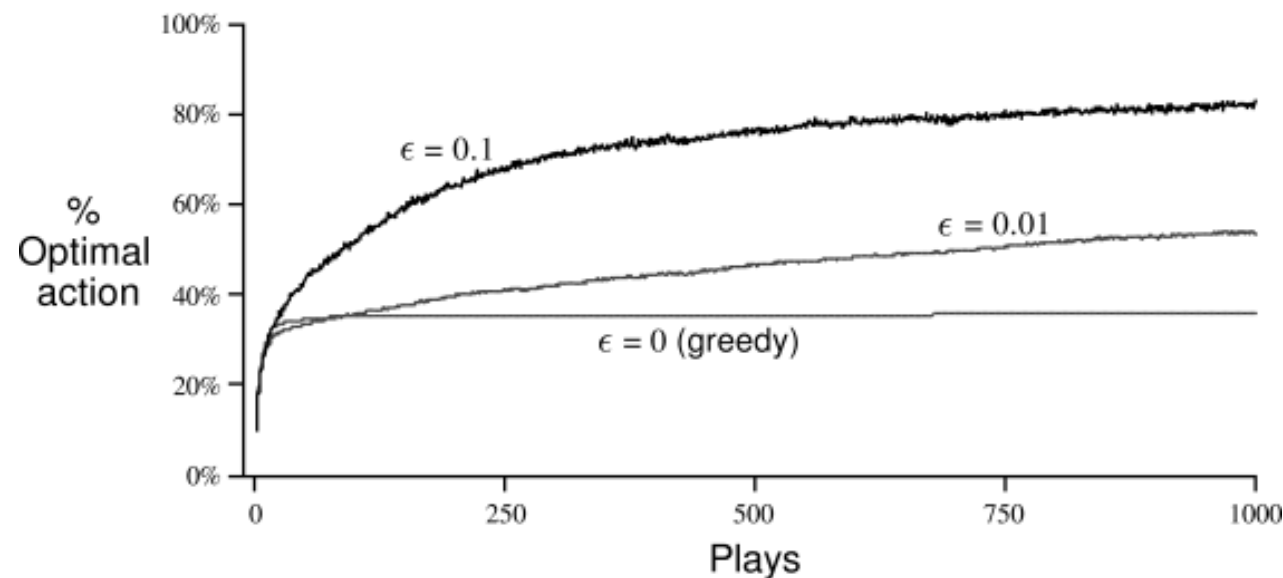
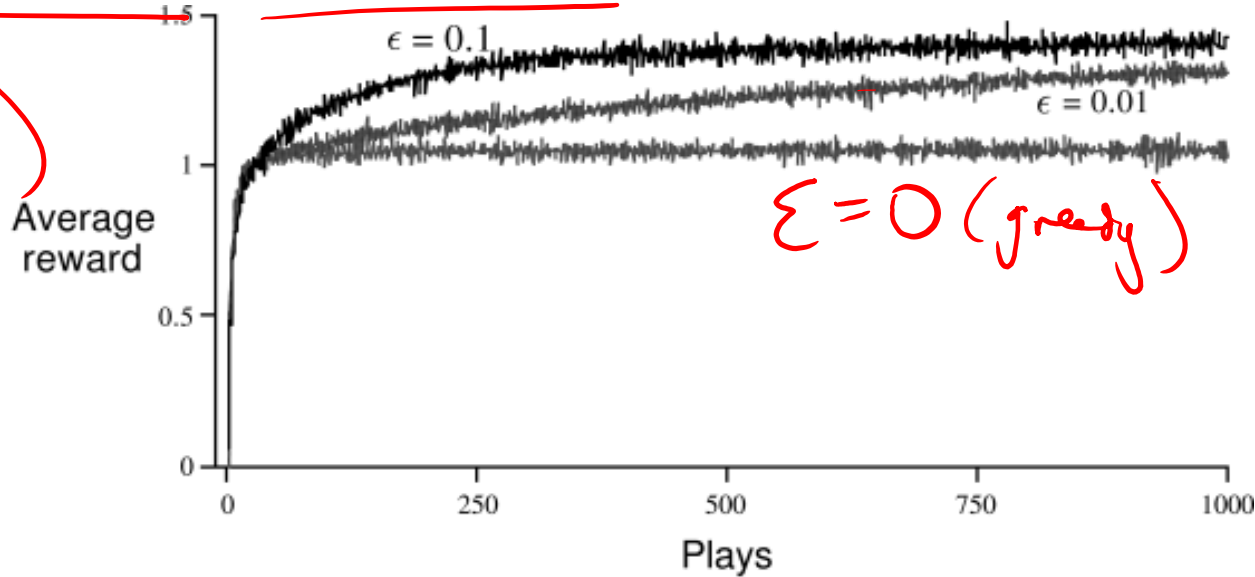
✓ ↓ ↓
 $N(0.1, 1)$ $N(-0.2, 1)$ $N(1.5, 1)$

$$s \sim N(a, b)$$

$$r \sim N(0, 1)$$

$$s = a + br$$

random, random(
 $\sim N(0, 1)$



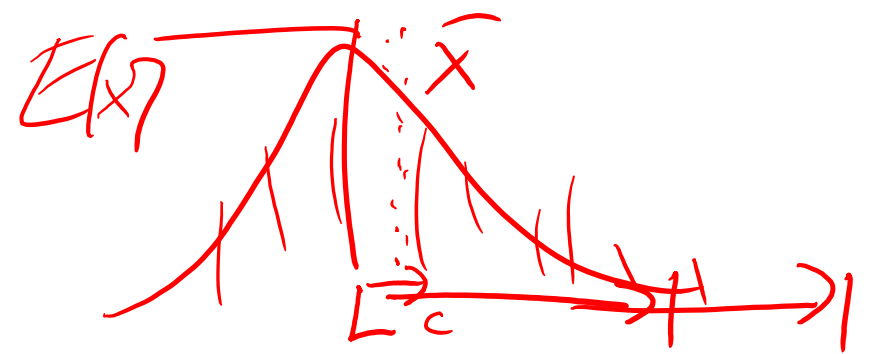
Problems?

$\epsilon?$

Boltzmann (Softmax) Exploration

$$\Pr(a) = \frac{e^{\beta \hat{Q}(a)}}{\sum_{a' \in A} e^{\beta \hat{Q}(a')}}$$

Chernoff-Hoeffding Inequality



- Let X be a random variable in the range $[0, 1]$ and $x_1, x_2, \dots, x_n$ be n independent and identically distributed samples of X . - $r \in$
- Let $\bar{X} = \frac{1}{n} \sum_i x_i$ (the empirical average) $\hat{Q}(a)$
- Then we have $P(\bar{X} \geq \mathbb{E}[X] + c) \leq e^{-2nc^2}$

Some fun math!

Handwritten notes in red:

$\mathbb{E}[X] \xrightarrow{c} \bar{X}$ 99% 95% $\delta = 0.05$

- $P(\bar{X} \geq \mathbb{E}[X] + c) \leq e^{-2nc^2}$
- Typically, we want to pick some kind of high confidence $1 - \delta$ such that we are very confident about our sample mean being close to the true expectation.
- If we want

$$P(\bar{X} \geq \mathbb{E}[X] + c) \leq \delta$$

What is c in terms of δ ?

Handwritten notes in red:

$\delta = e^{-2nc^2}$ 0.01

$$c = \sqrt{\frac{\log \delta}{-2n}}$$

More math

- We can pick δ to be whatever we want, so let's pick
- If we select $\delta = t^{-4}$

What is c ?

Handwritten notes and calculations:

$\leftarrow c \rightarrow$
 $E[X] \quad \frac{1}{X}$
 $t \rightarrow \infty \quad \frac{1}{t^4}$
 $\delta \rightarrow 0$

$$C = \sqrt{\frac{\log \delta}{-2n}} = \sqrt{\frac{\log t^{-4}}{-2n}} = \sqrt{\frac{-4 \log t}{-2n}}$$

Result boxed:

$$= \sqrt{\frac{2 \log t}{n}}$$

UCB1 (UCB = Upper Confidence Bound)

Key Idea: Optimism in the face of uncertainty

- Play each action once to get initial averages of arm values
- Keep track of counts of pulls for each arm n_i
- At each step t , select $\arg \max_i \bar{X}_i + c(i, t)$

- Where $c(i, t) = \sqrt{\frac{2 \cdot \log(t)}{n_i}}$

$t =$ total # actions

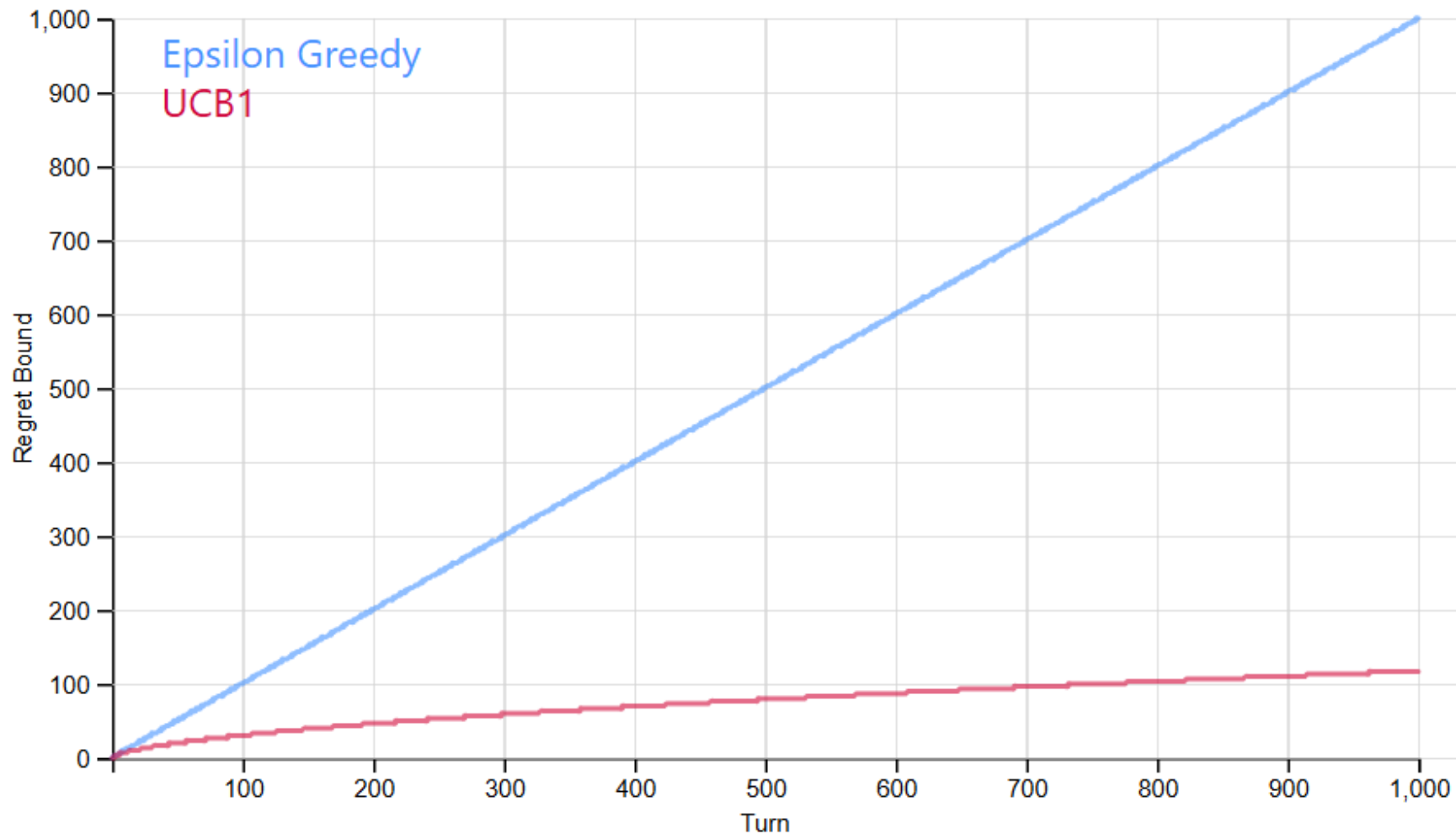
$t = \infty$



Regret

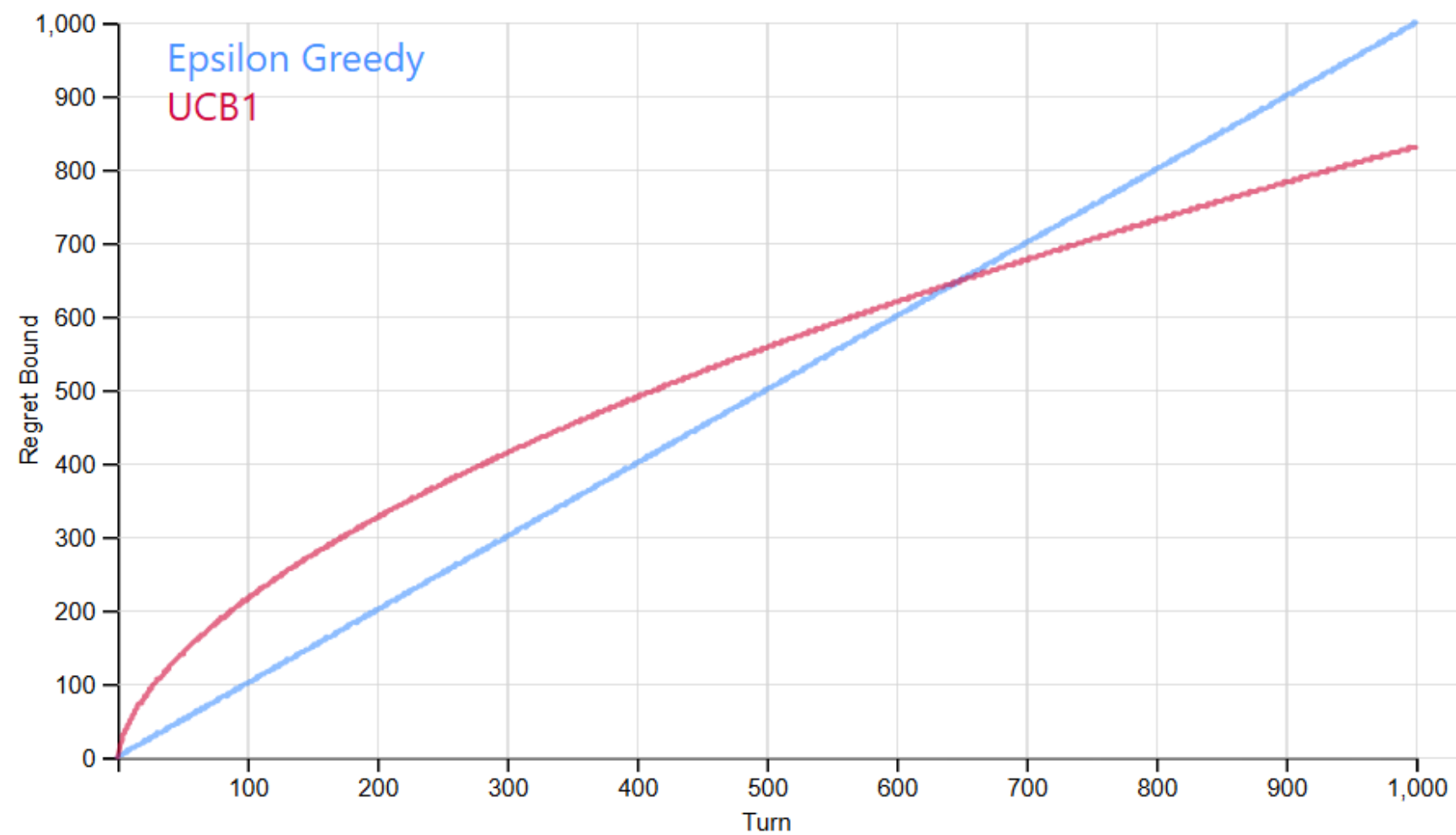
- Define μ^* as the maximum expected payoff over all k arms
- $\text{Regret}(T) = T\mu^* - \sum_{t=1}^T r_t$
- Epsilon-Greedy Regret
 - $O(T)$
- UCB1 Regret
 - $O(\sqrt{kT \log(T)})$
- A **No-Regret** algorithm is such that $\text{Regret}(T)/T \rightarrow 0$ as $T \rightarrow \infty$
 - Average regret goes to zero

Regret Bound vs. Turn



k (number of arms): T (number of steps):

Regret Bound vs. Turn



k (number of arms): T (number of steps):

Notes

- The version we derived is for rewards in range $[0,1]$
- What happens if rewards are in range $[a,b]$?
 - Just scale the upper confidence value

$$c(i, t) = (b - a) \sqrt{\frac{2 \cdot \log(t)}{n_i}}$$

In practice the scaling term is often just treated as a hyperparameter that controls exploration vs. exploitation.

$$c(i, t) = \alpha \sqrt{\frac{\log(t)}{n_i}}$$

Contextual Bandits

- **High-level definition**

At each round t :

- Observe **context** x_t
- Choose action $a_t \in \{1, \dots, K\}$
- Observe reward $r_t(a_t, x_t)$

Each arm has a context-dependent reward function:

$$\mathbb{E}[r \mid x, a] = f_a(x)$$

LinUCB

- **Assumption**

- For each arm a :
- Reward is linear in features
- Separate parameter vector per arm

$$\mathbb{E}[r \mid x, a] = x^\top \theta_a$$

- How should we choose arms (actions) given a context vector x ?
 - We maintain one linear regression model per arm
 - Plus uncertainty (e.g. upper confidence bound)

Aside Linear Regression

- Model assumption

$$y = x^\top \theta + \varepsilon$$

- Closed form solution

$$\hat{\theta} = (X^\top X)^{-1} X^\top y$$

Ridge Regression (adds L2 regularization)

$$\hat{\theta} = \arg \min_{\theta} (\|X\theta - y\|^2 + \lambda \|\theta\|^2)$$

- Also has a closed form solution:

$$\hat{\theta} = (X^{\top} X + \lambda I)^{-1} X^{\top} y$$

- How would you solve this in an online way? E.g., you only get one sample (x, y) at each timestep

LinUCB

- Optimism in the face of uncertainty in ***function space***.

LinUCB = online ridge regression + optimism via confidence ellipsoids

LinUCB Algorithm

Assumption

For each arm $a \in \{1, \dots, K\}$:

$$\mathbb{E}[r \mid x, a] = x^\top \theta_a$$

For rounds $t = 1, 2, \dots$

Observe context

$$x_t \in \mathbb{R}^d$$

For each arm a :

$$\hat{\theta}_a = A_a^{-1} b_a$$

$$\text{UCB}_a(x_t) = x_t^\top \hat{\theta}_a + \alpha \sqrt{x_t^\top A_a^{-1} x_t}$$

Select arm

$$a_t = \arg \max_a \text{UCB}_a(x_t)$$

Initialization

For each arm a :

$$A_a \leftarrow \lambda I_d \quad b_a \leftarrow 0_d$$

Observe reward

$$r_t \in \mathbb{R}$$

Update (chosen arm only)

$$A_{a_t} \leftarrow A_{a_t} + x_t x_t^\top$$

$$b_{a_t} \leftarrow b_{a_t} + r_t x_t$$

LinUCB

- LinUCB puts confidence ellipsoids on models!
- Works really well in practice!
- Very data efficient.
- Has been used heavily in production systems (ads, recommendations, etc)

Modern, Deep Learning Approaches

- Neural Contextual Bandits
 - Replace linear model with neural network $r \approx f_{\theta}(x, a)$
 - Train online from feedback
 - Figuring out good exploration/uncertainty is challenging
- Uncertainty via approximations
 - Ensembles
 - NeuralUCB (linear UCB on last-layer features)
 - Key idea: Use deep learning for representation, classical bandits for exploration.

Transitioning to Reinforcement Learning

- At what point does a contextual bandit become full reinforcement learning?